Lesson 5: Best-Fit Lines, Outliers, and Graph Quality

Unit

Working Like a Physicist: Collecting, Processing, and Presenting Data

Lesson focus

Students complete and improve their line graphs, learn how to draw a best-fit line, and identify possible outliers.

Assumptions

Students collected ramp practical data in Lesson 4.

Students have already plotted or started plotting their line graphs.

Learning objectives

By the end of the lesson, students should be able to:

1. Draw a suitable best-fit line.
2. Explain why dot-to-dot lines are usually not appropriate for experimental data.
3. Identify possible outliers.
4. Explain what should be done when an outlier is found.
5. Improve the quality of a line graph using a checklist.
6. Write a short conclusion from a graph.

Key vocabulary

Keyword	Student-friendly definition
Best-fit line	A line that shows the overall pattern of the results
Dot-to-dot line	A line drawn by joining every point directly to the next point
Outlier	A result that does not fit the pattern
Random error	Unpredictable variation between repeats
Systematic error	A repeated error that shifts all results in the same way

Equipment and resources

- Students' ramp practical results from Lesson 4
- Graph paper
- Rulers
- Pencils
- Example graphs, either printed or projected
- Graph quality checklist

Starter: Which graph is better?

Show three example graphs:

1. A dot-to-dot line.
2. A reasonable best-fit straight line.
3. A reasonable best-fit curve.

Ask students:

- Which graph best shows the pattern?
- What is wrong with joining dot to dot?
- Does a best-fit line need to pass through every point?

Main teaching: Best-fit lines

Explain:

A best-fit line should show the overall pattern of the results. It does not need to pass through every point.

For a straight-line relationship, students should aim to draw a line with roughly balanced points above and below the line.

Important message:

> *Experimental results are rarely perfect. The best-fit line shows the trend, not every single measurement.*

Main teaching: Outliers

Introduce:

An outlier is a result that does not fit the pattern of the other results.

Important message:

> *We should not automatically ignore an outlier. First, we should check whether there was a mistake or repeat the measurement if possible.*

Possible causes of outliers:

- incorrect measurement
- recording error
- equipment problem
- inconsistent release method
- misreading the scale

Activity 1: Spot the outlier

Give students this example dataset:

Ramp height / cm	Mean distance / cm
5	20
10	37
15	52
20	96
25	80

Students should notice that the value at 20 cm may be an outlier because it does not fit the general trend.

Discussion prompts:

1. Which value looks unusual?
2. Why does it look unusual?
3. Should we ignore it immediately?
4. What should we do first?

Expected answer:

The 20 cm result looks unusual. We should check the method and repeat the measurement before deciding whether to ignore it.

Activity 2: Finish the graph from Lesson 4

Students finish their line graph from the ramp practical.

They must include:

- title
- x-axis label and unit
- y-axis label and unit
- sensible scale
- accurately plotted points
- best-fit line
- possible outlier circled or commented on

Activity 3: Peer review checklist

Students swap books and check each other's graphs.

Graph feature	Yes/No
Title included	
x-axis labeled with unit	
y-axis labeled with unit	
Sensible scale	
Points plotted accurately	
Best-fit line drawn	
Dot-to-dot line avoided	
Outlier considered	

Written conclusion

Students write a short conclusion using this structure:

As ramp height increased, the distance travelled by the trolley __________.
I know this because the graph shows __________.
One possible source of random error was __________.
One result that may be an outlier is __________ because __________.

Plenary

Ask students to answer:

1. What is a best-fit line?
2. Why should we avoid dot-to-dot lines?
3. What is an outlier?
4. What should we do before ignoring an outlier?

Possible prep

Students are given a printed line graph with mistakes. They must identify and correct the errors, for example missing units, poor scale, dot-to-dot line, and unrecognized outlier.

Teacher notes

This lesson is a good opportunity to emphasize scientific judgment. Students often want a simple rule such as “ignore the weird result.” Encourage them to say, “repeat it first if possible.”